# Mathematics Reference: Ratio and Proportion

Educational Series

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## 1 Ratio

A ratio is a comparison of two quantities of the same kind by division. The ratio of $a$ to $b$ is written as $a : b$ or $\frac{a}{b}$.

- $a$ is called the **antecedent**.
- $b$ is called the **consequent**.

### 1.1 Types of Ratios

1. **Duplicate Ratio:** The duplicate ratio of $a : b$ is $a^2 : b^2$.
2. **Sub-duplicate Ratio:** The sub-duplicate ratio of $a : b$ is $\sqrt{a} : \sqrt{b}$.
3. **Triplicate Ratio:** The triplicate ratio of $a : b$ is $a^3 : b^3$.
4. **Compounded Ratio:** The compounded ratio of $(a : b)$ and $(c : d)$ is $(ac : bd)$.

## 2 Proportion

An equality of two ratios is called a proportion. If $a : b = c : d$, then $a, b, c, d$ are said to be in proportion, written as:

$$a : b :: c : d$$

In this expression:

- $a$ and $d$ are called the **extremes**.
- $b$ and $c$ are called the **means**.
- **Rule:** Product of Extremes = Product of Means ($ad = bc$).

### 2.1 Mean Proportion

If $a, b, c$ are in continued proportion ($a : b = b : c$), then $b$ is the mean proportion between $a$ and $c$.

$$b^2 = ac \implies b = \sqrt{ac}$$

## 3 Properties of Proportion

If $\frac{a}{b} = \frac{c}{d}$, then the following properties hold:

- **Invertendo:** $\frac{b}{a} = \frac{d}{c}$
- **Alternendo:** $\frac{a}{c} = \frac{b}{d}$
- **Componendo:** $\frac{a+b}{b} = \frac{c+d}{d}$
- **Dividendo:** $\frac{a-b}{b} = \frac{c-d}{d}$
- **Componendo and Dividendo:** $\frac{a+b}{a-b} = \frac{c+d}{c-d}$
- **Addendo:** $\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$

## 4 Variations

- **Direct Variation:** $x \propto y \implies x = ky$ (where $k$ is a constant).
- **Inverse Variation:** $x \propto \frac{1}{y} \implies xy = k$.

## 5 Solved Example

**Problem:** If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.

**Solution:** To combine the ratios, we make the term for $B$ equal in both ratios. The L.C.M. of 3 and 4 is 12.

- $A : B = 2 : 3 = (2 \times 4) : (3 \times 4) = 8 : 12$
- $B : C = 4 : 5 = (4 \times 3) : (5 \times 3) = 12 : 15$

Therefore, $A : B : C = 8 : 12 : 15$.